Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

| Proposer | Name | Freek Bomhof | Email | Freek.bomhof@tno.nl |
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| Institute Name | TNO | | | |
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## 2. Technology Overview

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| Title of Proposed Technology | GPT-NL: Creating a Large Language Model that is fully lawful and transparent | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Information & Communication | | Knowledge services | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Software | Software solutions | R&D / Engineering services | Other R&D / Engineering-related Technologies |
| Concept of Proposed Technology | Artificial Intelligence, specifically Large Language Models (LLMs) have become the core of many innovations and it has also become the core of criticism: models that are based on stolen data, trained on questionable data, and supplied in a non-transparent way. This is a problem for organizations that want to deploy Responsible AI in high-risk situations.<br>TNO has created GPT-NL that effectively addresses this criticism. The GPT-NL approach is the most comprehensive approach towards transparent, trustworthy and responsible AI. GPT-NL's unique and distinctive features are:<br>- The training data is only based on truly open data (creative commons license), on fully licensed data (with an agreement in place), and on synthetic data (based on sources like WikiData) | | | |

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| | - A complete data curation pipeline is used to filter out inadmissible content, for instance, to respect the privacy of any persons that are not public persons<br>- The model is trained with attention to bias (e.g., respecting underrepresented groups) and toxic language |
| **Rationale for Joint R&D Project** | GPT-NL has been started in the Dutch context, with partners (the national supercomputing centre and the Dutch Forensics Institute). The initial version has been running for 2 years on a budget of 13.5M€.<br>The aim of GPT-NL has been to prove that it is actually possible to build a performing, safe, reliable and lawful LLM. The proof has been delivered for the Dutch context and language, and we are now eager to extend and test the approach. This includes other contexts: currently the model is aimed at summarization, simplification and Q&A (RAG) applications. We expect to extend the GPT-NL approach to other languages, because we believe that our approach is not only valuable in the Dutch context.<br>Outside the scope of the current proposal: there are concrete plans to add multimodality, too: speech (sound) and data (tabular, time series), to create a multimodal foundational model. |
| **Key Words** | 1. Data collection and curation<br>2. Data provider agreements<br>3. Large Language Model pretraining and finetuning<br>4. Evaluation of the performance of Large Language Models |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level 5 | End of the Project | Level 7 |
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| R&D Objectives | The project is probably best carried out in two phases.<br><br>The objective for the first phase would be to validate the core assumption. This assumption is that a Korean version of GPT-NL would be a viable exploitable product for a Korean SME. The objective of the first phase is to validate GPT-NL's unique approach and features for the Korean context.<br><br>After validation, key R&D objectives for the second phase will be:<br>- How can the existing successful approach of GPT-NL be translated to the Korean context?<br>- How and to what extent is the state of the art GPT-NL data curation pipeline useful for the Korean language as well? | | | |
| R&D Target Specifications | The end result of the project will be an LLM:<br>- Trained on curated data<br>- Has a size of ~27B parameters<br>- Performance comparable to GPT-NL<br>- Evaluation for three key tasks: summarization, simplification, and Q&A (RAG)<br>However, the exact specifcations will be outcome of the analysis phase. | | | |
| R&D Plan | We propose the following structure but are open for alternative R&D plans.<br><br>Phase 1:<br>Analysis of the R&D goal. The features of the GPT-NL product would have to be compared to the requirements for the Korean situation. What is the use case or product idea that the Korean SME would like to pursue, and what would the specification of this result consequently have to be?<br><br>Phase 2:<br>Once a decision has been made to build a Korean LLM, the below proposal could be an outline with very tentative durations. Note that the actual plan depends on the specific more detailed analysis of the situation.<br>1) Data acquisition & licensing (6 months)<br>2) Data curation (2 months)<br>3) Training infrastructure (4 months)<br>4) Pretraining (3 months)<br>5) Finetuning dataset creation / acquisition (3 months)<br>6) Finetuning (2 months)<br>7) Evaluation (2 months)<br>The above work packages can be overlapping. | | | |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | The collaboration will be set up as a joint project, where TNO and the Korean SME will jointly build the end result. TNO will bring in the experiences and methodologies that have been developed while creating GPT-NL and will make a proposal for the project structure, tasks/work packages, results and milestones. This will be agreed with the Korean SME before start of the project.<br><br>There will be a project steering board with senior representatives from both partners. During the project execution, regular project management meetings will take place to take decisions and monitor progress. For each work package, regular technical meetings will be organized. These meetings will normally take place online. At regular intervals, physical meetings (either in the Netherlands or South Korea) will take place in order to execute joint sprint development cycles. These face to face meetings also help to promote mutual trust and understanding. A joint document and code repository will be set in place.<br><br>From the TNO side, the project will be supported by a small team of professionals that have experience Generative AI, and specifically in the development of GPT-NL. This includes data collection & curation expertise, training & finetuning, cybersecurity, ethical, and project management expertise. The team consists of highly esteemed scientific researchers (post doctorate, professors) and experienced engineers and architects.<br><br>To summarize: the Research institute (TNO) will be responsible for:<br>- Supplying and adapting the methodologies and software that is needed to train the model<br>- To jointly evaluate the data curation pipeline and adapt it to the Korean context and language<br>- To design, monitor and oversee the actual training (pretraining and finetuning) process<br>The compute may be subcontracted and managed by TNO, but this may also be transferred to the Korean SME partner, depending on the analysis of the requirements and the possibilities. |
| **Expected R&R of Korean SME Partner** | All specific Korean conditions and aspects are best executed by the Korean SME. This includes the legal context, and the collection of training data, including the license agreements with data providers. Obviously, TNO will provide advice and consultancy for these tasks based on their experience with these processes in the Dutch context.<br><br>The Korean SME will be the partner that will benefit from the local exploitation of the project resuilt. Therefore, the testing and evaluation of the trained model is best executed by the Korean SME: not only because of knowledge of the Korean language and cultural context, but also because it enables the Korean SME a thorough understanding of the project result, which is necessary for succesful exploitation.<br><br>To summarize, the Korean SME partner will be responsible for:<br>- Analyzing the specific legal conditions (for instance, privacy, copyright) in South Korea & ensuring compliance<br>- The collection of training data and finetune data, including the license agreements<br>- The implementation of the training infrastructure, including the cybersecurity measures *(depending on the possibilities, this activity could also be carried out by TNO)*<br>- The testing and evaluation of the trained model |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | The project outcome will be that South Korea can use a sovereign Large Language Model, which completely adheres to legal conditions in Korea, and which is fully transparent and trustworthy. This model enables Korea to use Generative AI on their own terms.<br><br>Expected impacts will be:<br>• the Korean SME that engages in this project will have had a similar learning experience as the GPT-NL team in the Netherlands: how to build a lawful and reliable Large Language Model that contributes to digital sovereignty.<br>• South Korea has taken an additional step forward in digital sovereignty. While tailored Korean LLMs that are trained from scratch already exist (such as A.X 3.1), the GPT-NL-inspired data curation and approach to lawfully and fairly use data, will be a valuable additional experience. |

## 6. References

(1) GPT-NL: Towards a Public Interest Large Language Model

(2) T. Barbereau en L. Dom, "GPT-NL: Towards a Public Interest Large Language Model," Dutch Organization for Applied Scientific Research (TNO), Den Haag, 2023. Available: https://publications.tno.nl/publication/34643373/5aTCTsaH/barbereau-2023-gpt.pdf

(3) T. Barbereau en L. Dom, "GPT-NL: Towards a Public Interest Large Language Model," *Proceedings of the Public Interest AI Workshop (PIAI 2024)*, CEUR-WS, vol. 3958, pp. 1–6, 2024. Available: https://ceur-ws.org/Vol-3958/piai24-short2.pdf

**Freek Bomhof** (MSc) Freek Bomhof graduated in Pattern Recognition (University of Twente) and joined KPN Research in various positions around the area of digitization (researcher, project manager, programme manager, R&D manager). He joined TNO in 2003 and has had several positions there in various application domains, mostly in the Safety & Security sector. His interest is responsible AI & Data: Trustworthy AI and the application of Privacy Enhancing Technologies for secure analysis of sensitive data. He has set up and chairs the Dutch National Innovation Centre for Privacy Enhancing Technologies. He has set up and coordinated various EU R&I projects, and is involved in many TNO projects, amongst other GPT-NL. He is now programme director of the TNO-wide AI research programme Appl.AI (8 year programme, ~10M€ per year) and he chairs the TNO AI Guild, bringing together the community of AI professionals within TNO. Freek is member of the Board of Directors within BDVA (Big Data Value Association) and also leads the BDVA taskforce on policy & regulation issues.

**Dr. Rer. Nat. Julio A. de Oliveira Filho** is Innovator Scientist and Engineer of safe AI systems at TNO. I research, design, and engineer the next generation of intelligent autonomous systems—spanning robotics, quantum networks, IoT sensor systems, automotive technologies, and modern Defense applications. As the leading architect of GPT-NL, the first Dutch-English large language model trained fully from scratch, and one of the creators of NetSquid, the world's most widely used quantum network simulator, I thrive at the intersection of advanced AI, complex systems, and cutting-edge engineering.

My work focuses on developing adaptive, distributed, embedded systems through rigorous model-driven engineering, machine learning, and high-impact automation tooling. I'm passionate about elevating AI engineering

into a mature discipline—where autonomous systems are not only powerful, but reliable, safe, and explainable by design.

**Saskia Lensink** is a senior consultant and business developer and specializes in language and speech technologies. She applies her knowledge of NLP and ASR in various projects, and is active in a diverse set of consortia and networks to promote sovereign and high-performing European large language models. She is currently the product owner of GPT-NL

Saskia has a background in linguistics. She earned her PhD in Leiden University. As a linguist, she examines the structures within language. She measures how the human brain, ears, and speech apparatus work. She also researches how this can be captured in computer models (natural language processing and computational linguistics). As a product owner at GPT-NL, Saskia brings the various teams and the external world together. This includes data acquisition, system architecture, and data curation, as well as the requirements and interests of external stakeholders.

**Maaike de Boer** (PhD) is a senior Scientist at TNO within the Data Science department. She has a background in AI and linguistics (Msc 2013, PhD 2017). At TNO Maaike mainly focuses on text mining / natural language processing (NLP) and hybrid AI: combining data-driven methods and knowledge-driven methods (such as ontologies) to get the best of both worlds. Projects include scientific projects and European or Horizon2020 projects such as TrustLLM and Cyclopes, but also Dutch and B2B projects such as the Growth Fund project Vaardig met Vaardigheid and Norm Engineering. Her projects are mainly focused in the defence, labour market and health domains. In previous years she has worked on NLP techniques for strategic foresight. Further details can be found through LinkedIn and Google Scholar.

**Stephan Raaijmakers** is a senior scientist at TNO, specialized in machine learning-based natural language processing (NLP), and a professor of Communicative AI at Leiden University, where he focuses on responsible use of Large Language Models in communicative user interfaces. He is the lead scientist of GRAIL, a TNO initiative for Responsible Generative AI. He has ample Horizon Europe experience, a.o. as the Technical Coordinator of two Horizon Europe projects. Stephan has authored two books on deep learning-based NLP and LLMs. In recent years, he has contributed to TNO projects on machine learning-based trend analysis of textual data.

**Roos Bakker** (MSc) Roos Bakker (MSc) is a data scientist at TNO within the Data Science department, and a PhD student at Leiden University. She obtained an MSc in Artificial Intelligence from Utrecht University in 2018 and a BA in linguistics in 2015. In 2022, she started a PhD in collaboration with TNO and Leiden University on the topic of Knowledge Graph Extraction, Enrichment, and Evaluation. Roos is experienced in manual ontology development and Natural Language Processing, and is passionate about combining both to advance the field of knowledge representation and extraction. She is also ardent about developing code in an open and structured manner, examples can be found in open-source repositories such as Normative systems · GitLab, Covid-19 Ontology on Test Information, Common Greenhouse Ontology . Further details can be found at https://www.linkedin.com/in/roos-bakker and Roos Bakker - Google Scholar

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |

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